MAKRAND GHODKE

M.Sc. Student, AI/ML and Software Engineering

makrandghodke02@gmail.com Darmstadt, Germany

Open to working student and HiWi roles in ML, AI and data; up to 20 hrs/week during the semester

linkedin.com/in/makrand-ghodke github.com/MakrandGhodke

Summary

M.Sc. student in Information & Communication Engineering at TU Darmstadt (AI/ML specialization). I build and test software around machine-learning models: a PyTorch and CLIP image-classification pipeline, a FastAPI backend in Docker on Render, and 38 pytest tests with GitHub Actions CI. I like finding out why a model gives the wrong answer, and I once took one from 30% to 100% accuracy by tracing a silent preprocessing bug.

Projects

SpendWise — Household Finance Tracker (PWA) 03/2026 – 05/2026

- Built an OCR-based receipt ingestion pipeline into a PostgreSQL schema (Supabase) with per-user data isolation (Row Level Security), plus a real-time 6-month analytics dashboard
- Implemented a REST API with transaction CRUD (filtering, sorting, pagination) via Next.js API routes, deployed as a PWA on Vercel
- Tested the app as a standalone PWA on both iOS and Android

DigiGuide — AI Landmark Recognition App 10/2025 – 02/2026

- Built and evaluated an image-classification pipeline in a 4-person team: CLIP with FAISS similarity search, served through a FastAPI endpoint
- Diagnosed a silent data-preprocessing failure (missing RGB conversion) that held accuracy at 30% with no errors thrown. Traced it stage by stage and rebuilt the preprocessing, lifting accuracy from 30% to 100% across all 5 target classes

DigiGuide Backend — Deployment, Testing & CI, *github.com/MakrandGhodke/digiguide* 10/2026

- Containerized the FastAPI and CLIP backend with Docker and deployed it on Render; fixed an out-of-memory failure by pre-caching the model in the Docker build
- Wrote a pytest suite (38 tests) with the model and an external API mocked, so it runs in seconds without network access. The tests found three defects (path traversal, a 500 error on invalid uploads, a hard-coded JWT secret), fixed with a regression test for each so they cannot come back
- Set up GitHub Actions workflows: one runs the test suite on every push and pull request, another builds the Flutter iOS app

Education

M.Sc. Information and Communication Engineering, *Technische Universität Darmstadt* 10/2024 – Present
4th semester, AI/ML specialization Courses: Introduction to AI · Data Science 1 & 2 · Explainable and Ethical AI Darmstadt

B.Tech Electronics and Telecommunication, *Bharati Vidyapeeth Deemed University* 11/2020 – 07/2023
CGPA: 9.06 / 10 Pune

Professional Experience

Web Developer Intern, *Exposys Data Labs* 07/2022 – 08/2022
Built and tested responsive web interfaces across four browsers (Chrome, Firefox, Safari, Edge) as part of a 5-member Agile/Scrum team

Operations Support Technician, *Flextronics Technologies* 07/2021 – 12/2021
Streamlined production data reporting by redesigning Excel-based tracking tools, reducing manual entry errors and improving cross-team visibility

Technical Skills

ML & Vision
Python, PyTorch, CLIP, FAISS, scikit-learn, pandas, NumPy, Computer Vision, Model Evaluation

Backend & Delivery
FastAPI, REST APIs, Docker (familiar), Render, Vercel, GitHub Actions (CI), pytest, Git, GitHub, Agile/Scrum

Web & Data
TypeScript, Next.js, React, Flutter/Dart, Node.js, SQL, PostgreSQL (Supabase), Tailwind CSS

Programming
Python, TypeScript, JavaScript, Java, C++, SQL

Tools
Jupyter Notebook, VS Code, AI assistants (Claude, Antigravity)

Languages

English
C1 (Fluent)

German
A2–B1 (Basic, actively improving)

Hindi
Native

Marathi
Native